SETHL: PARTIALLY SPECIFIED HAND ARTICULATION CONTROL FOR VIDEO DIFFUSION

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ABSTRACT

Pose-conditioned video generators assume complete trajectories, although users may specify only important fingers or intervals. We formulate *partially specified hand control*: obey declared constraints while preserving variation elsewhere. SetHL extends Hand Labanotation (HL) from symbols to distributions over unspecified cells. A masked transformer completes missing symbols, a certified in-cell sampler provides continuous variation without changing HL labels, and a kinematic layer drives rank-16 LoRA-adapted Wan2.1-Fun-Control 1.3B. On WLASL, three-seed experiments show that hard HL improves specified-symbol accuracy by 2.78 points and reduces angular error by 2.44° versus continuous control, but reduces localized diversity by 1.68°. SetHL recovers 1.15° of that diversity while retaining significant accuracy, leakage, and detection gains over continuous control. Its hypervolume gain is not significant, but conditional denoising MSE improves by 2.98%. The results support structured partial constraints as an editable video-diffusion interface that separates controlled from free hand articulation.

***Index Terms*—** controllable video generation, hand motion, Hand Labanotation, diffusion models

1. INTRODUCTION

Hand motion is both semantically dense and difficult to synthesize. Many pose-based controllers consume a complete joint trajectory, while sparse-control methods usually sparsify time or landmarks rather than letting authors declare which articulations are unconstrained [1, 2, 3, 4]. Complete trajectories are suitable for reenactment, but form a poor authoring interface: a user may require an index finger to point inward during one interval while leaving the remaining articulation unspecified. Treating missing coordinates as zeros or hallucinating a single completed skeleton confuses "free to vary" with "fixed to an arbitrary estimate."

Hand Labanotation (HL) [5] provides a useful alternative. It describes each of 20 hand-local bone directions by one of 26 interpretable spatial symbols. The original work used HL for recognition, documentation, and approximate one-shot skeleton recovery. We ask a different question: can a partial symbolic program define a *set* of valid videos rather than one trajectory?

We propose SetHL-Wan. Specified HL symbols are preserved exactly, whereas a temporal completion network represents each unspecified cell by a categorical posterior. Samples are forward-kinematically spatialized and lifted to the control-latent interface of Wan [6]. Only low-rank adapters [7], the completion network, and the lifting module are trained. Crucially, no dense joint coordinates bypass the symbolic stream.

Our contributions are: (1) a task definition and metric that jointly measures constraint compliance and diversity localized to unconstrained hand parts; (2) a probabilistic extension of HL with certified symbol-preserving continuous samples, coupled to a differentiable kinematic Wan controller; and (3) matched-budget tests against continuous and learned-token alternatives, including structured missing-control and noise stress tests.

2. RELATED WORK

Controllable video diffusion. ControlNet [1] established spatial conditioning for diffusion models, while SparseCtrl [2] studied temporally sparse video conditions. Recent work conditions video models on pose or trajectories [3, 4, 8, 9, 10]. Sparse 3-D hand control has also addressed occlusion and geometric fidelity [11], but "sparse" there describes landmarks or missing observations, not author-declared degrees of freedom. These methods seek one accurate motion trace; our target is selective constraint with localized residual variation. Learned 3-D motion and mesh tokens avoid rasterized controls [12, 13], but encode complete motion with task-specific vocabularies rather than an interpretable, partially specified program.

Hand and sign generation. WLASL [14] enables large-scale word-level sign recognition. Recent generators improve sign fidelity through pose adapters [15] or compressed multimodal tokens [16]. They motivate stronger hand evaluation, but do not expose a symbolic partial-control interface.

Symbolic hand motion. HL [5] quantizes hand-local directions and is robust to small pose noise. We retain its fixed

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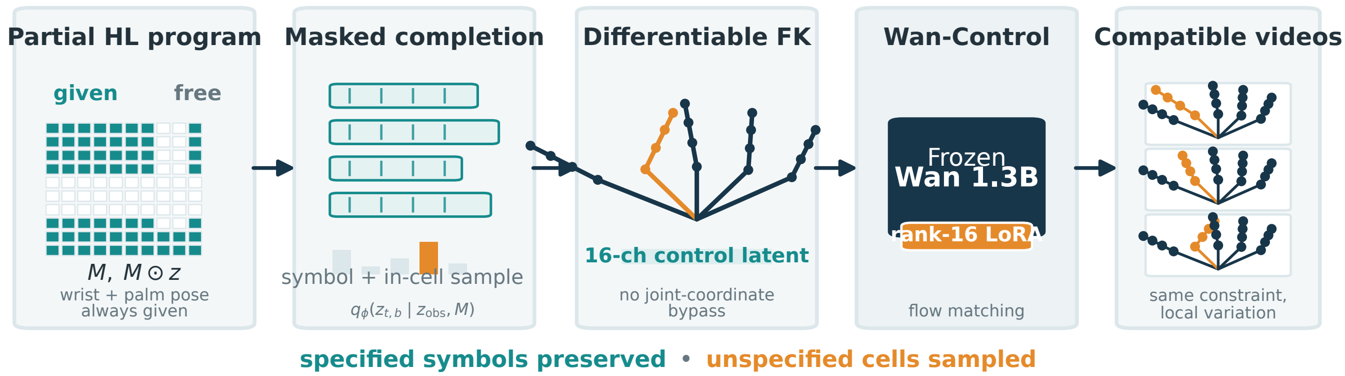

Fig. 1. SetHL treats a partial hand instruction as a set rather than a single completed skeleton. Given HL cells are copied exactly; a masked transformer predicts distributions for free cells, which are sampled and spatialized by forward kinematics before conditioning a LoRA-adapted Wan model. Continuous within-cell samples preserve their selected HL symbol by construction.

semantics, but replace deterministic cell-center recovery with a posterior over unspecified cells and train it as a generative condition. A learned 26-entry VQ codebook is included to test whether gains come merely from discretization.

3. METHOD

Let $X = (x_1, \dots, x_T)$ be a video and $J_t \in \mathbb{R}^{2\times21\times3}$ its two-hand landmarks. The control mask $M \in \{0,1\}^{T\times2\times20}$ indicates which bone-time cells the user specifies.

3.1. Partial Hand-Labanotation programs

For each hand we construct the palm frame R_t from the wrist, index metacarpophalangeal joint, middle metacarpophalangeal joint, and little-finger metacarpophalangeal joint as in [5]. A bone direction $d_{t,b}$ is mapped into this frame and assigned to its nearest member of the fixed 26-direction codebook $C = \{c_k\}_{k=1}^{26}$:

$$z_{t,b} = \arg\max_k \langle R_t^\top d_{t,b}, c_k \rangle. \tag{1}$$

The input program contains $(M, M \odot z)$, wrist trajectories, palm frames, and reference bone lengths. It never contains per-joint image coordinates. Specified cells are one-hot, preventing continuous sub-cell information from leaking through soft labels. We interpret each symbol as its spherical Voronoi cell rather than only its center.

For missing cells, a transformer q_ϕ attends jointly over time, hands, and the 20-bone trees:

$$\begin{aligned} q_{t,b} &= q_\phi(z_{\text{obs}}, M), \\ \mathcal{L}_{\text{HL}} &= -\log q_{t,b}(z_{t,b}) + \lambda_h [1 - \langle \bar{c}_{t,b}, c_{z_{t,b}} \rangle]. \end{aligned} \tag{2}$$

where $\bar{c}$ is the normalized posterior mean. The second term respects the geometry of nearby HL regions instead of treating all symbol mistakes as equally severe.

3.2. Kinematic control-latent lifting

A hard straight-through categorical sample first selects a center c_z. For unit $v \perp c_z$, we draw

$$\tilde{d} = \cos\alpha\, c_z + \sin\alpha\, v, \quad \alpha < \tfrac{1}{2} \min_{j \neq z} \arccos(c_z^\top c_j). \tag{3}$$

By the spherical triangle inequality, c_z remains the nearest codeword. This gives continuous variation with a certificate that the HL instruction is unchanged. Forward kinematics reconstructs bone endpoints from wrist position, palm frame, and bone length. Bone tokens are bilinearly splatted along their 2-D segments and decoded by three small 3-D convolutions into a 16-channel latent $H_\psi(P)$ at Wan's temporal and spatial resolution. We concatenate it with the first-frame latent, matching the native Wan-Control input contract.

Given clean latent v, noise ϵ, and flow-matching interpolation $v_\sigma = (1-\sigma)v + \sigma\epsilon$, we optimize

$$\mathcal{L} = \| f_{\theta,\Delta}(v_\sigma, \sigma, H_\psi(P)) - (\epsilon - v) \|_2^2 + 0.1\mathcal{L}_{\text{HL}}, \tag{4}$$

where the 1.3B Wan backbone θ is frozen and Δ denotes rank-16 LoRA parameters. At inference, independent posterior samples alter the control trajectory only where the program permits it, while diffusion noise supplies appearance variation.

4. EXPERIMENTS

Data and protocol. We extract one 17-frame clip from each WLASL video with reliable MediaPipe Hands [17] tracks.

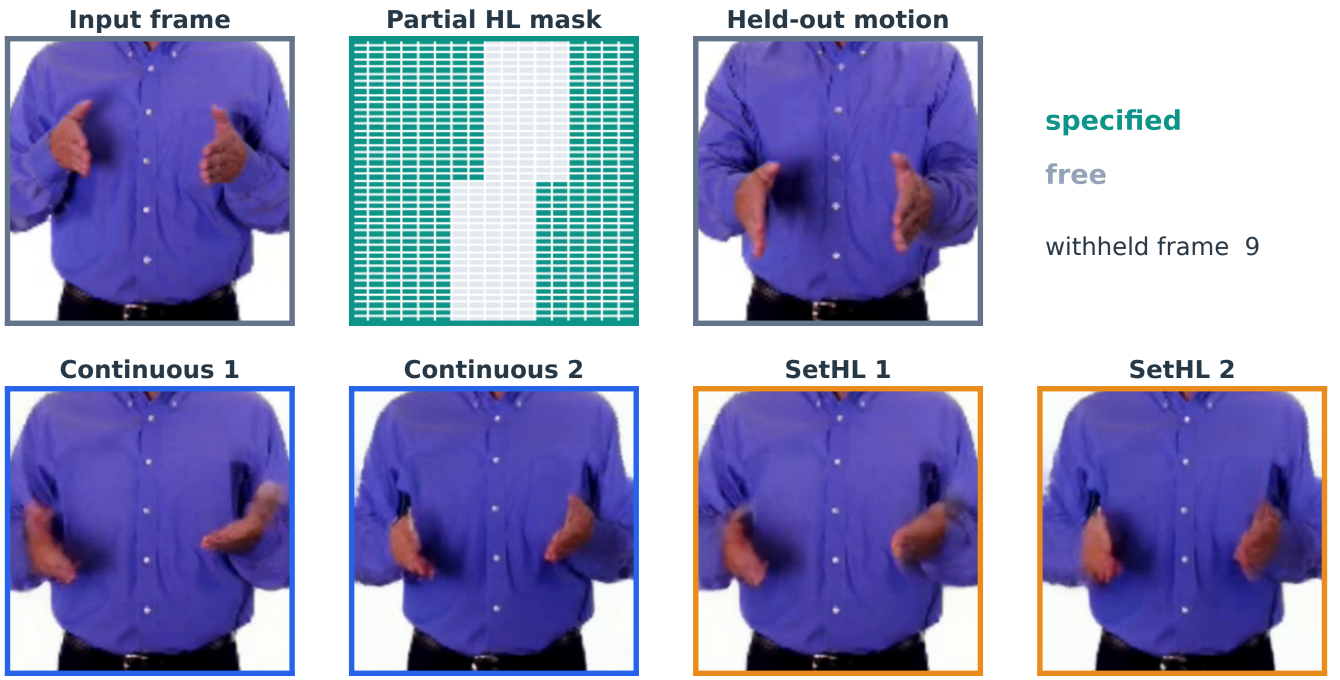

Fig. 2. Generated-video comparison on the source at the median three-seed SetHL hypervolume (a deterministic selection rule). Teal mask cells are specified and gray cells are free; frames are taken at the center of the withheld interval. Columns show paired diffusion seeds under continuous and SetHL control.

Because different WLASL identifiers can decode to identical frames, we hash the decoded RGB tensors and retain one record per hash, prioritizing test, then validation, then train without consulting any outcome. This removes 193 duplicate records and yields 2,059 content-unique clips in a 1,402/392/265 train/validation/test partition; no RGB hash crosses a split. Frames are resized to 256^2. Every learned method uses the same Wan checkpoint, rank-16 LoRA, 3,000 updates, and three seeds. SetHL trains 21.85M parameters in total (LoRA plus its completion and lifting modules), while the matched continuous model trains 21.84M. With cached Wan-VAE latents, each 3,000-step seed takes about 28 minutes on one Ascend 910B2C NPU. Text and CLIP embeddings are fixed to zero for every method so the comparison isolates first-frame and motion-control conditioning. We select $\lambda_h = 1$ from $\{0, 0.25, 1\}$ using validation data only. We likewise select posterior temperature 2 from $\{1, 1.5, 2\}$ on eight fixed validation sources and share it across all probabilistic categorical controls.

Baselines and metrics. We compare continuous Gaussian directions, a training-set 26-entry VQ codebook, hard HL centers, SetHL without the spherical loss, and full SetHL. Continuous and raster controls retain exact observed geometry and therefore serve as information-richer upper bounds, while parameter counts and training updates remain matched. Training samples masks that remove a distal joint set, one complete finger, or a contiguous temporal interval. Standalone completion averages all three; the generated-video benchmark uses the predeclared interval mask. We report specified-symbol accuracy, specified angular error, hidden-cell angular diversity, specified-cell symbol disagreement, hand detection, temporal acceleration error, and denoising MSE. The primary endpoint is the Pareto hypervolume dominated by compliance–localized-diversity operating points (reference $(0, 0)$) obtained by scaling the control latent. Localized diversity multiplies hidden-cell angular diversity by semantic agreement on specified cells and hand-detection rate. Thus semantic leakage and untrackable hands are penalized while legal angular variation inside an HL cell is not. Confidence intervals use 10,000 paired bootstrap resamples of source videos. Only the predeclared hypervolume test is confirmatory; secondary intervals are pointwise and unadjusted for multiplicity. Generated-video metrics use all 25 test clips meeting the fixed criterion that both hands are tracked in at least 90% of source frames. As an evaluator ceiling, re-encoding these real clips and rerunning the same tracker gives 92.94% hand detection, 80.26% HL agreement, and 9.06° angular error against the archived pseudo-labels. As a post-hoc quality diagnostic, we also measure source-conditioned cosine similarity in the penultimate feature space of a Kinetics-400-pretrained R3D-18 [18]; this metric was not used for model selection.

Main results. Table 1 reports the locked comparison. The paired SetHL–continuous difference in the primary end-

Table 1. Interval partial-control results on the WLASL test split. Higher is better for accuracy, localized diversity, and hypervolume; lower is better for angle. Values average three training seeds; bold marks the best 26-symbol method, and paired 95% intervals are reported in text.

Method	Acc.(%)↑	Ang.° ↓	LocDiv° ↑	HV↑
Information-rich controls				
Raster skeleton	56.0	26.32	10.84	9.49
Continuous Gauss.	50.5	31.27	11.02	8.37
26-symbol controls				
Learned VQ-26	51.2	30.44	**11.57**	**8.88**
Hard HL	**53.3**	**28.83**	9.34	8.10
SetHL w/o sph. loss	51.4	30.94	11.04	8.61
SetHL-Wan	52.6	29.62	10.49	8.50

point is 0.13 (95% CI [-0.55, 0.78]); against VQ-26 it is -0.37 ([-1.13, 0.27]). The predeclared HV-superiority criterion is therefore not met. Against the no-spherical-loss ablation, SetHL changes accuracy by 1.16 points ([-0.26, 2.44]), angle by -1.32° ([-2.76, 0.06]), and HV by -0.10 ([-0.71, 0.48]). Relative to continuous control, SetHL changes specified accuracy by 2.03 percentage points ([0.27, 3.82]), angular error by -1.65° ([-3.40, 0.14]), leakage by -3.77 points ([-6.53, -0.95]), and detection by 4.03 points ([2.03, 6.11]); its localized-diversity difference is -0.53° ([-1.74, 0.75]). Against learned VQ-26, SetHL changes specified accuracy by 1.33 percentage points ([-0.17, 2.99]) and leakage by -1.55 points ([-3.90, 0.82]); neither interval excludes zero. Hard HL makes the tradeoff especially clear: versus continuous control it improves accuracy by 2.78 points ([1.20, 4.52]), angular error by -2.44° ([-4.03, -0.84]), and leakage by -6.16 points ([-9.06, -3.17]), while losing -1.68° localized diversity ([-3.12, -0.28]). SetHL recovers 1.15° of localized diversity over hard HL ([0.21, 2.17]), while its compliance accuracy and angular-error differences from hard HL are not significant; its specified-cell disagreement increases by 2.40 points (95% CI [0.84, 3.97]). Under fully specified control, SetHL accuracy is 54.0% versus 51.9% for continuous control (paired difference 2.15 points, 95% CI [1.00, 3.37]). On all 265 held-out clips, interval-conditioned denoising MSE is 0.1031 for SetHL and 0.1062 for continuous control, a 2.98% reduction; the paired change is -0.00316 (95% CI [-0.00384, -0.00250]). R3D feature similarity is 0.9360 versus 0.9308 (paired change 0.0052, 95% CI [0.0032, 0.0072]), indicating no loss of source-level video consistency alongside the semantic gains.

Representation analysis. On held-out directions, fixed HL has 16.93° mean quantization error, versus 16.26° for VQ-26. Under coordinate noise $\sigma = 0.005$, however, HL retains 58.6% of symbols compared with 58.0% for VQ. This separates geometric precision from stability and tests whether a learned vocabulary alone explains the result.

5. CONCLUSION

Partial hand instructions should define constraints, not prematurely collapse generation to one completed skeleton. SetHL extends a fixed, interpretable notation into a distribution-valued controller for Wan. Its value is tested by where variation occurs, not merely by whether generated motion is diverse. The present evaluation is limited to short, hand-focused WLASL clips and tracker-derived pseudo-labels; broader domains and human judgments remain future work.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study used only the publicly available WLASL dataset and involved no new data collection or interaction with human participants; no ethical approval was required.

7. ACKNOWLEDGMENT

OpenAI Codex assisted with language drafting throughout the manuscript, implementation of evaluation and statistical-analysis code, and scripting of Figs. 1–2. The authors reviewed all outputs and take responsibility for the study design, paper content, and reported results.

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